Food Delivery Time Prediction

From exploratory analysis to model evaluation: predicting delivery duration in minutes



Problem Statement

Distance (r = 0.78) and prep time (r = 0.31) drive baseline delays, plus weather.

2

Core Variables

Regression cuts average error from 17.5 to 6.05 min and reveals key drivers.

4

ML Solution

Inaccurate Estimates

1

Baseline averages miss by ~17.5 min per order, negatively impacting customer satisfaction

External Friction

3

High traffic raises median delivery time to ≈65 min, vs ≈51 min when traffic is low.

Dataset Overview

1,000 food delivery orders, with the goal of predicting delivery time

1,000
orders in the dataset

9
raw columns

80/20
train / test split (800 / 200 orders)

30
missing values in each of 4 columns



Target: Delivery_Time_min

Features: distance, weather, traffic level, time of day, vehicle type, preparation time, courier experience

Order_ID is dropped: a unique identifier with no predictive value.

Delivery Time at a Glance

Average delivery takes about 57 minutes, with a long tail up to 153

56.7

mean (min)

55.5

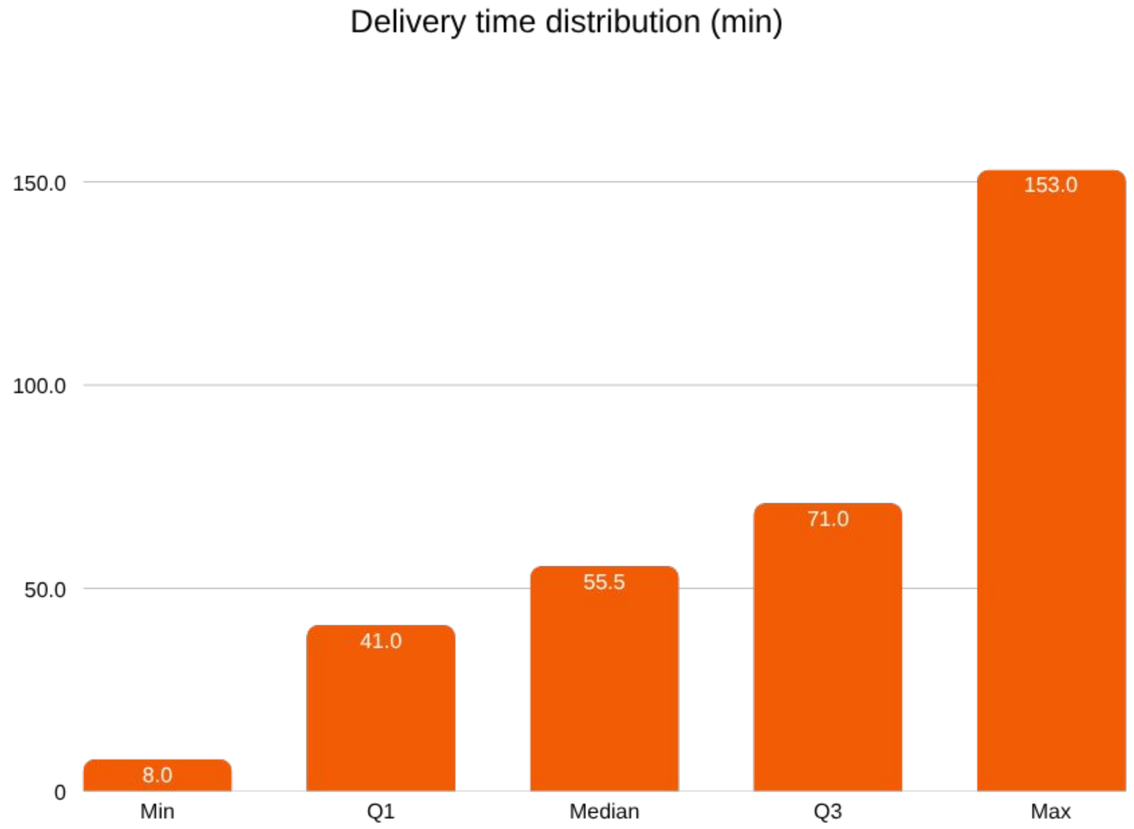
median (min)

22.1

standard deviation

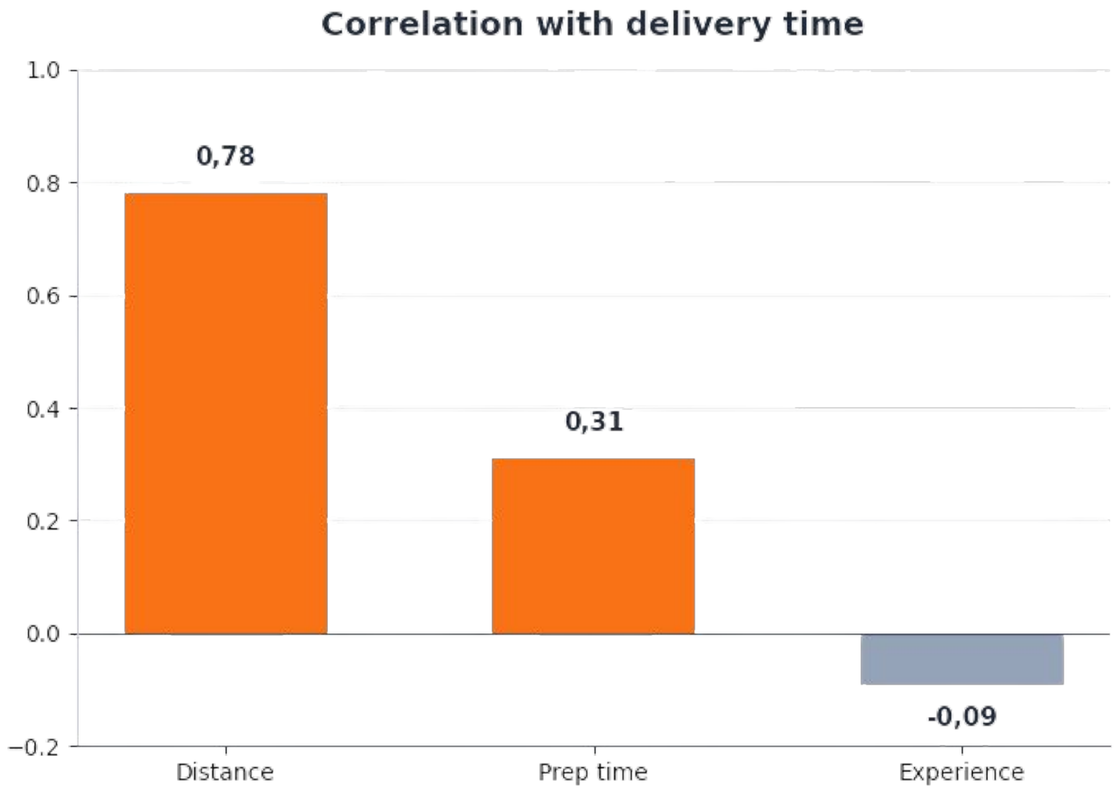
153

longest delivery (min)



What Drives Delivery Time

Distance is by far the strongest signal



Distance: r = 0.78, so longer trips take clearly longer.

Conditions: Snowy weather and high traffic have the highest medians (≈66 and ≈65 min vs ≈52 and ≈51 for Clear and Low).

Little effect: vehicle type and time of day barely shift delivery time.

Data Preparation

Split first, then clean using training statistics only, to prevent data leakage

1

Split first

80/20 split (800 train, 200 test, random_state = 42) before any cleaning, so test data never influences preprocessing.

2

Impute missing values

Train-only statistics: mode for the 3 categorical columns, median for courier experience. 30 gaps per column down to 0.

3

Check outliers

IQR check on distance, preparation time and courier experience found 0 outliers, so no capping was needed.

Feature Engineering

Four new features built from input columns only, with no use of the target

Feature	Definition	Purpose
Traffic_Ord	Low, Medium, High → 0, 1, 2	Ordered traffic scale
Dist_x_Traffic	Distance × Traffic_Ord	Load from distance and congestion
Dist_per_Exp	Distance ÷ (Experience + 1)	Distance relative to courier skill
Bad_Weather	1 if Rainy, Snowy or Foggy	Flags adverse conditions

16

final model features

One-hot encoding

Weather, Time_of_Day and Vehicle_Type; test columns aligned to train.

Modeling Approach

Four regression models compared on the same train and test split

Baseline

Predicts the training mean: the bar every model must beat.

Linear Regression

Simple and interpretable, with one weight per feature.

Random Forest

200 trees; captures non-linear patterns.

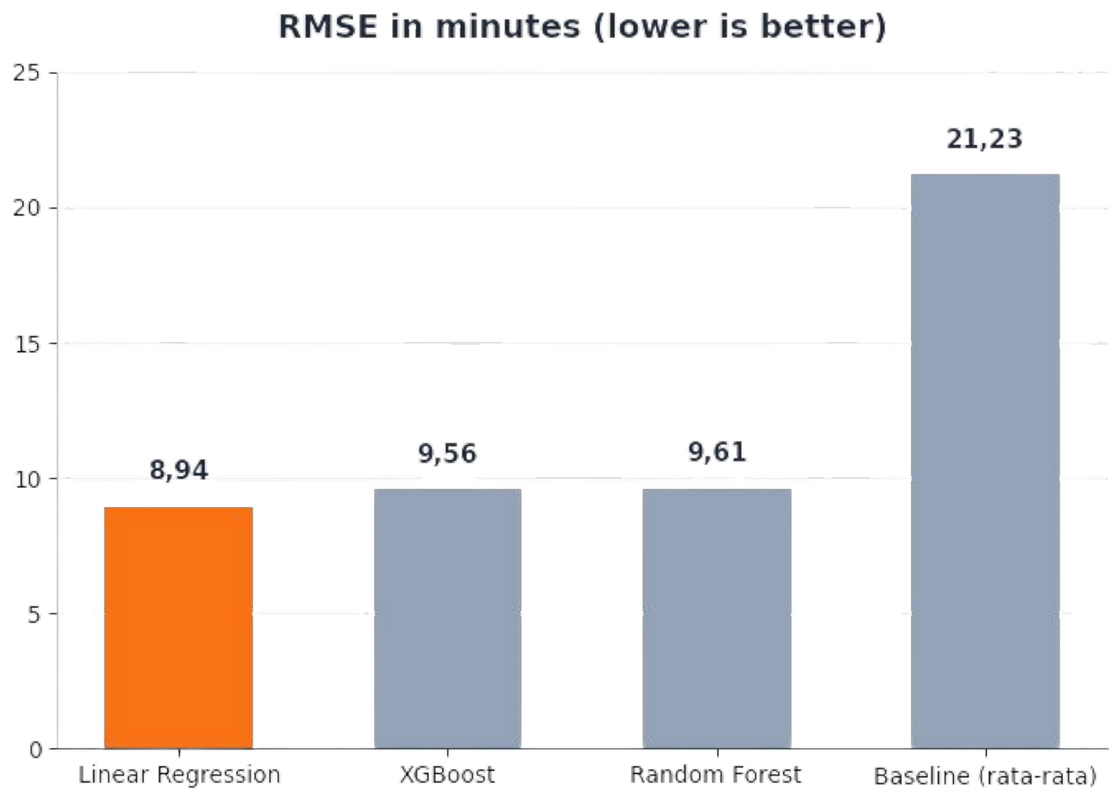
XGBoost

200 boosted trees, learning rate 0.05, max depth 3.

Evaluated on 200 held-out orders with MAE, RMSE and R^2

Model Results

Linear Regression wins: lowest error and R^2 of 0.822

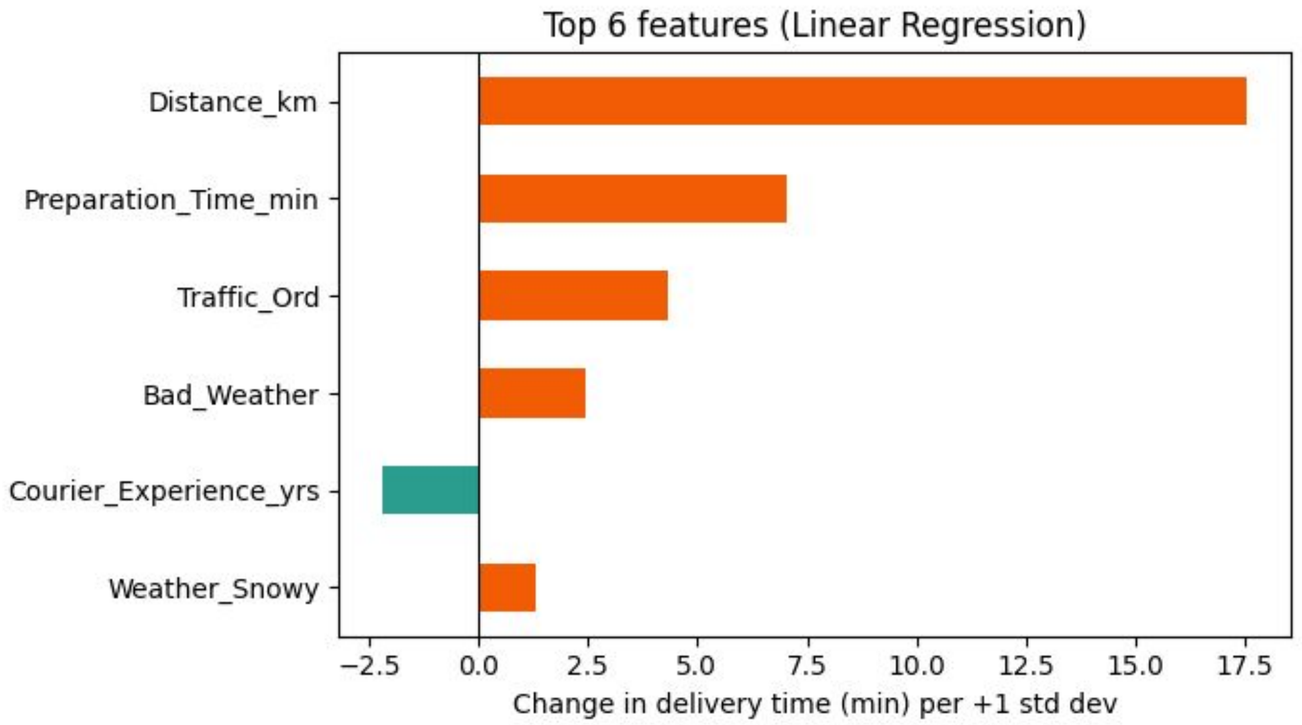


Model	MAE	RMSE	R^2
Linear Regression	**6.05**	**8.94**	**0.822**
XGBoost	6.66	9.56	0.796
Random Forest	6.80	9.61	0.794
Baseline (mean)	17.50	21.23	-0.006

MAE and RMSE are in minutes. A baseline R^2 near 0 means the mean alone explains none of the variation.

Feature Importance

Linear Regression shows the same story: distance and preparation time lead



+17.6 min

Distance, per +1 std dev

+7.0 min

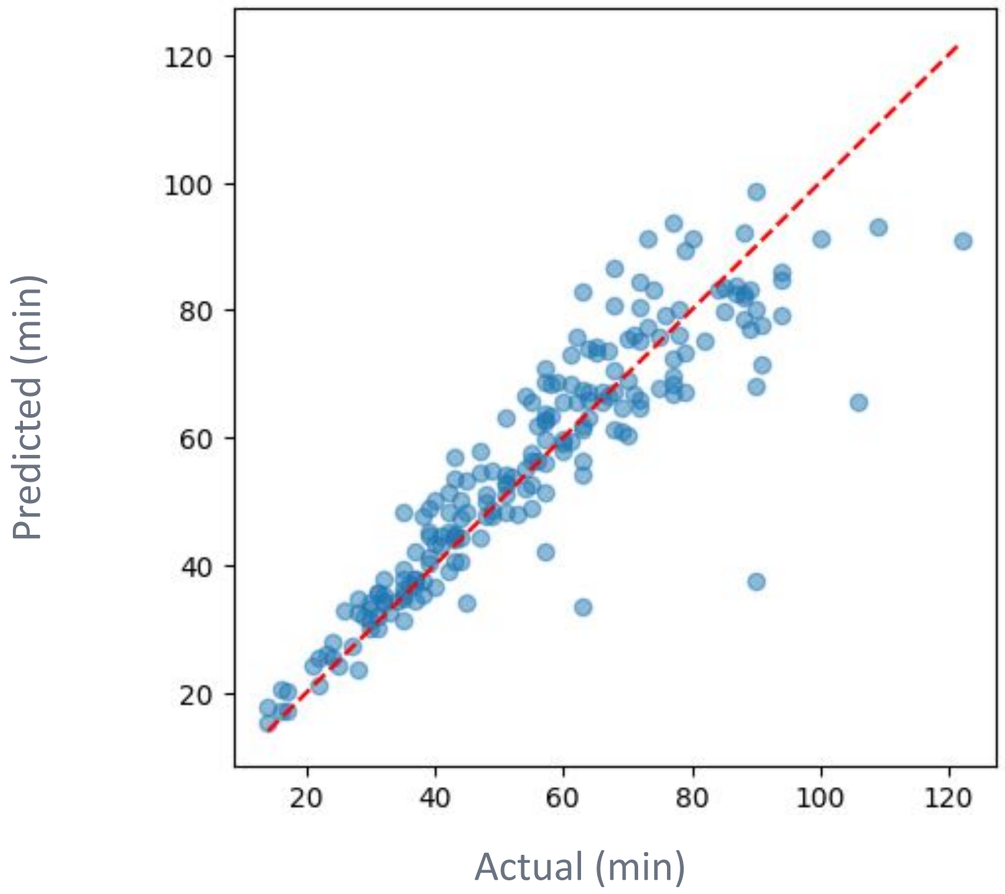
Preparation time, per +1 std dev

Change in delivery time (min) per +1 standard deviation of the feature

1 std dev of distance ≈ 5.73 km

Actual vs Predicted

Linear Regression predictions track real delivery times closely



On the line: most points sit close to the ideal red dashed line.

Typical error: MAE of 6.05 min, RMSE of 8.94 min.

Watch out: a few long deliveries (around 90 min) are clearly under-predicted.

Sample Predictions

Applying the saved preprocessing and the Linear Regression model to new orders

Dist. (km)	Weather	Traffic	Time of day	Vehicle	Prep (min)	Experience (yrs)	Predicted (min)
5.0	Clear	Low	Morning	Bike	10	3.0	**28.9**
12.5	Rainy	High	Evening	Scooter	20	1.0	**78.2**
20.0	Foggy	Medium	Night	Car	25	5.0	**98.0**

The longest trip, with medium traffic and fog, gets the longest predicted time (98 min).

Insights & Recommendations

Insights

Distance dominates

Distance dominates: r = 0.78, and +17.6 min per one-std-dev increase in the model.

A simple model wins

Linear Regression: MAE 6.05 min, R^2 0.822, vs a 17.5 min baseline MAE.

Conditions add delay

High traffic (≈65 vs ≈51 min) and snowy weather (≈66 vs ≈52 min) raise medians.

Recommendations

Replace average-based ETAs

Use the model to cut typical estimate error from 17.5 to about 6 min.

Plan for risky orders

Add time buffers for long-distance, high-traffic and snowy deliveries.

Keep improving the model

Cross-validate, tune, and study under-predicted long deliveries (~90 min).

Documentation

Link Github

Food Delivery Time Prediction

Link Collab

Food Delivery Time Prediction

THANK YOU